# Bangladesh University of Engineering and Technology

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A Project Proposal on

# Sustainable Drainage System Improvement Plan for BUET Area

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## Introduction

Drainage systems play a vital role in maintaining the health, safety, and functionality of any urban environment. At BUET, frequent rainfall during the monsoon season often results in waterlogging across various parts of the campus. These issues disrupt student mobility, damage infrastructure, and create potential health hazards. Despite being a leading engineering institution, BUET still relies on an outdated and undersized drainage network that can no longer handle current or future demands.

This project has been undertaken to explore whether a more sustainable, long-term solution can be designed—one that not only addresses the immediate drainage problems but also supports environmental resilience and better campus functionality.

## Project Proposal

The existing drainage infrastructure at BUET was designed at a time when rainfall patterns, land use, and campus density were very different. Over the years, the system has become insufficient for present conditions.

BUET students face real, everyday problems because of the outdated and poorly maintained drainage system. During monsoon season or even after moderate rain, water collects across campus, making it difficult to get to places like the library, cafeteria, or even class. It affects our daily routine, safety, and quality of campus life.

One serious incident showed just how critical this issue is — a term final exam was delayed for almost a month after a contamination problem, caused by improper drainage where pipes for toilets and drinking water were connected.

Other issues contain:

- Waterlogging is regularly experienced during and after moderate rainfall.
- Drains are often clogged and poorly maintained.
- Academic schedules are interrupted.
- Health risks are increased due to stagnant water and mosquito breeding.
- No sustainable mechanism is currently in place to handle stormwater.

This project is needed not just for convenience, but for our health, safety, and a better campus experience overall. BUET, as a leading institution in engineering and innovation, must also reflect these values in how its campus infrastructure is managed. While academic advancements progress rapidly, the surrounding environment needs systems that are equally thoughtful and future-focused. Drainage is not just a utility concern—it is a vital part of sustainable campus development.

## Problems of Existing Drainage System in BUET:

**Frequent Waterlogging:** Waterlogging refers to a condition where excess water causes flood-like situation. The waterlogging problem is severe in the main campus and the southern side of the campus. In a low-lying area like BUET, when it rains heavily the existing drainage system cannot remove or absorb the water quickly. As a result, water enters in halls as well as in the campus of BUET. This causes much difficulty in the movement of students and stuffs, transportation and other activities. It also damages roads and other infrastructure which might cost a lot to repair. When the water enters in the hall, it damages the study materials, electronic devices and other useful things of the students who are living in the ground floor. At that time, they face much difficulties for their living which causes a great harm to their studies. If this happens during term final, the damage is irreparable.

**Severe Mosquito Infestation:** As a result of waterlogging, BUET faces a major outbreak of mosquito. Water remains accumulated in halls and campus area and so a large number of mosquitoes breed there. Open drains and filthy water serve as perfect breeding spots for mosquitoes which causes diseases like dengue, chikungunya etc. Students fell sick and cannot concentrate on their studies and other activities. Sometimes they also do not get proper treatment. Indirectly drainage system is causing a lot of problems to the students and also staffs.

**Coliform Detected:** Fecal coliform has been found in the water of BUET a few months ago which the students had been using. Fecal coliform is a subgroup of bacteria which is a warning sign of unsafe, unhygienic water and a public health risk. The breakdown of fecal coliform in the water of BUET was very sudden and it came to knowledge after causing severe health issues to the students like diarrhea, allergy, vomiting etc. Some students were hospitalized for using this contaminated water. To solve this problem, the chlorine level was increased in water for few days which caused skin irritation, hair loss etc to the students. All of these happened during the PL (Preparatory leave) which is considered the most important period for the students of BUET where they take preparation for their upcoming term final. The date of the term final was rescheduled until the problem of solved and it was postponed by almost one month.

**Disrupted Communication:** Due to the inadequate drainage system, waterlogging occurs and it not only affects the transportation but also disrupts communication among the students and staffs and also among the common people around BUET. The movement becomes difficult in academic buildings and halls. Sometimes the classes are cancelled and labs are rescheduled for this. Even the term final was also postponed once.

## Root Causes of the Problems of Drainage System:

**Blocked & Undersized Drainage Pipes:** The draining network is not organized in BUET. The system is outdated and the pipes were also designed long time ago. Due to urban growth and heavier rainfall, the volume of water has increased. But comparing to the increasing water volume, the size of the pipes has remained the same. Moreover, due to accumulated waste, mud, plastic-the pipes are blocked. These are not cleaned regularly. As a result, in the undersized pipes, blockage also occurs.

**Poor Maintenance & Damaged Networks:** Due to wear and tear, soil erosion, heavy traffic pressure, and lack of timely repairs, the drainage pipes are being damaged day by day highlighting the fact that drainage system in BUET lacks proper maintenance. In BUET, sometimes the water doesn't drain at all because of the cracked, broken pipes and collapsed network- causing leakage and insufficient flow. Drains not being cleaned regularly blocks the flow of water. When it rains heavily, the water overflows onto the halls and campus area which disrupts the movement and so on.

**Flat Slope:** BUET area is about 0.5-1% flat slope in respect to the near areas, causing waters back flow to the lower surface area of BUET. However having a flat slope causes slow or stagnant water movement. This leads to water accumulation in halls and campus area as the water cannot drain efficiently during rainy season. The velocity of water become slow due to the flat slope and gravity causing more dirt and mud in the pipes and worsens the drainage network.

**Lack of Proper Sewage Disposal:** Sewage disposal means the process of purifying wastewater. BUET lacks of a well-structured sewage system that can collect, separate, and treat black and grey water. Black water contains human waste which comes from toilet and grey water comes from sinks, showers and laundry being less harmful than black water. A well-furnished drainage system requires separate pipelines for sewage. Due to the lack of proper sewage, the black water had mixed with the using water of BUET which caused the fecal coliform related problem in BUET.

**Hydraulic Mismatch:** When the design capacity of a drainage system does not match with the actual volume of water, hydraulic mismatch occurs. In BUET campus and hall areas, hydraulic mismatch occurs due to undersized pipes, uneven flow rates, lack of slope, poor connectivity etc.

## Expected Outcomes:

**No Waterlogging:** By this project, the pipe size will be updated with the increasing volume of water, damages will be repaired and so on. This updated drainage system will prevent the accumulation of rainwater in halls and campus areas. More safe campus environment will be ensured by this for students.

**Proper Draining Facility:** A smooth and timely removal of excess water will be ensured by a well-designed drainage system where the pipes will be in an appropriate size, slopes will be improved, and a proper sewage system. There will be no risk of overflow, blockages and any kind of damages.

**Improved Sanitation:** There will be no mixing of black and grey water with the purified water as there will be a separate and well-maintained sewage disposal. This will ensure a cleaner campus where no students will get sick using the water, no mosquito breeding. It will ensure better hygiene and public health for the students.

## Project Management Lifecycle

The project has been categorized into 4 phases – from the beginning to the delivery.

1. **Phase-1 (Concept):**
    - Identify waterlogging hotspots on campus and adjacent areas.
    - Review existing drainage condition & feasibility study.
    - Explore alternative retrofit layouts.

2. **Phase-2 (Planning):**
    - Conduct topographic & hydrologic surveys
    - GIS mapping & digital hydraulic modelling (HEC-RAS)
    - Civil & hydraulic design of pipe network
    - Assess utility conflicts & risk mitigation plan
    - Finalize budget allocations & project brief
    - Secure approvals (RAJUK, LGED, BUET)

3. **Phase-3 (Execution):**
    - Procure pipes, manholes & equipment
    - Shift existing utilities & establish bypass lines
    - Excavation, pipe laying & alignment
    - Road resurfacing & reinstatement
    - On-site supervision, quality assurance & safety monitoring
    - Progress reporting & change-control

4. **Phase-4 (Transfer):**
    - Final inspections & hydraulic performance testing
    - Training & handover to DSCC/WASA maintenance team
    - Obtain formal sign-offs from BUET, DSCC, WASA & other stakeholders
    - Reconcile all contracts, close out procurements & finalize accounting

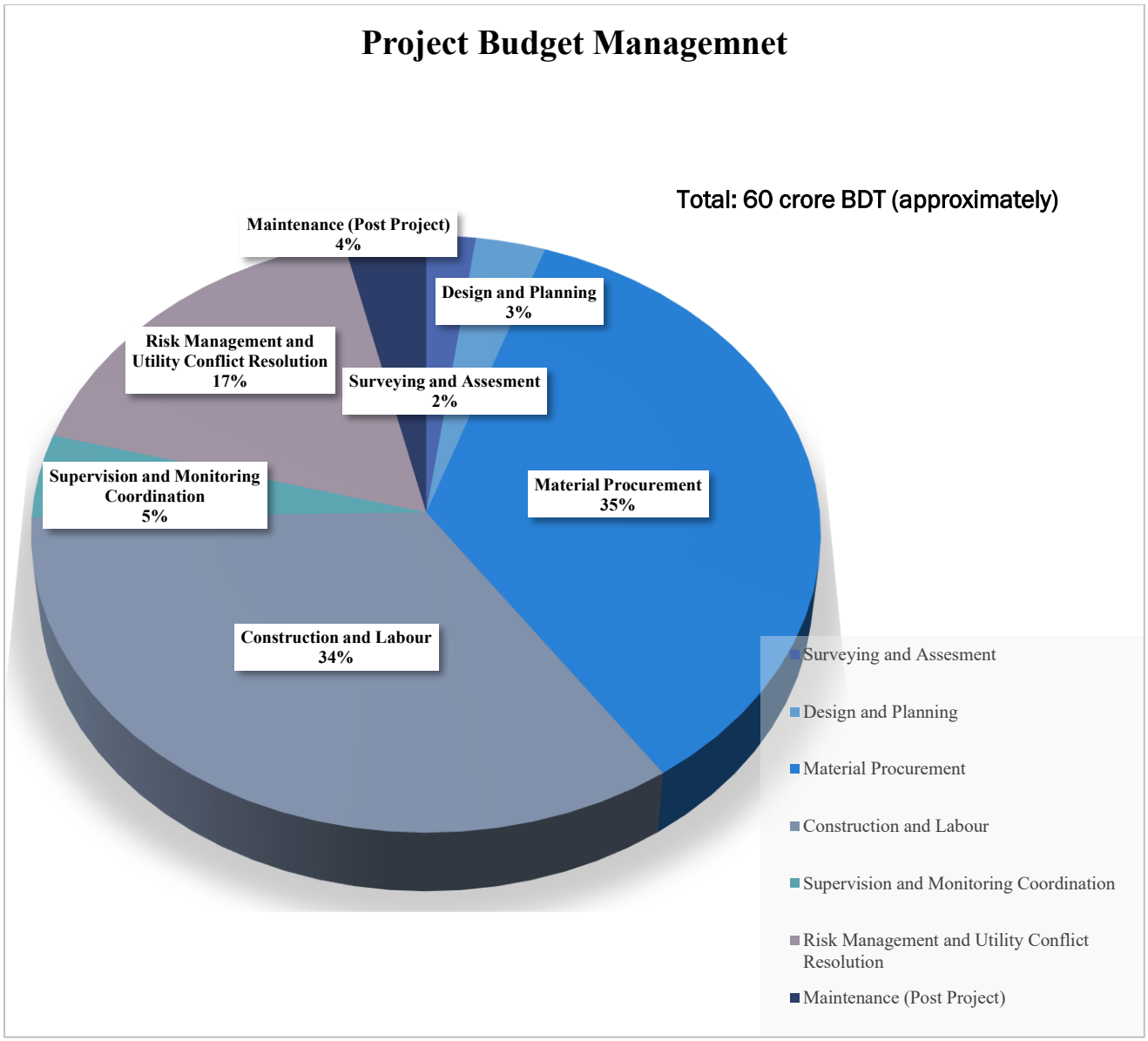
## Project Budget Management

The project costs an estimation of 60 crore BDT which covers all the task and subtask involved.

| Task | Cost |
|---|---|
| Surveying and Assessment | 1.2 crore BDT |
| Design and Planning | 1.7 crore BDT |
| Material Procurement | 19.5 crore BDT |
| Construction and Labour | 19 crore BDT |
| Supervision and Monitoring Coordination | 2.5 crore BDT |
| Risk Management of Utility Conflict | 9.5 crore BDT |
| Maintenance (Post-Project) | 2 crore BDT |
| **Total** | **55.4 ~ 60 crore BDT** |

| Task | Subtask | Budget | Subtotal |
|---|---|---|---|
| Surveying and Assessment | Topographic and hydrologic survey | 50 lakh BDT | 1.2 crore BDT |
| | Pipe network condition assessment | 30 lakh BDT | |
| | Geotechnical investigation | 30 lakh BDT | |
| | GIS mapping and digital modeling (HEC-RAS) | 10 lakh BDT | |
| Design and Planning | Civil and hydraulic design of the drainage system | 1.2 crore | 1.7 crore BDT |
| | Consultation fees (BUET professors, expert advisors) | 50 lakh BDT | |
| Material Procurement | Concrete drainage pipes (different diameters) | 10 crore BDT | 19.5 crore BDT |
| | Manhole covers | 50 lakh BDT | |
| | Construction-grade finer sand, coarse aggregates and cement | 5 crore BDT | |
| | Temporary bypass lines | 1 crore BDT | |
| | Pipe-jointing and alignment equipment | 2 crore BDT | |
| | Safety gear and PPE for construction workers | 1 crore BDT | |

| | | | |
|---|---|---|---|
| Construction and Labour | Excavation works | 5 crore BDT | **19 crore BDT** |
| | Pipe laying and alignment | 8 crore BDT | |
| | Road resurfacing after work | 4 crore BDT | |
| | Skilled and unskilled labor wages | 2 crore BDT | |
| Supervision and Monitoring Coordination | Field supervision teams | 2 crore BDT | 2.5 crore BDT |
| | Administration and coordination | 50 lakh | |
| Risk Management of Utility Conflict | Shifting existing underground utilities (gas, electricity) | 8 crore BDT | 9.5 crore BDT |
| | Emergency incident fund (for damages or accidents) | 1 crore BDT | |
| | Site safety and environmental mitigation | 50 lakh BDT | |
| Maintenance (Post-Project) | Scheduled cleaning and inspections | 1 crore BDT | 2 crore BDT |
| | Minor repair work and staffing | 1 crore BDT | |
| **Total = 55.4 crore BDT ~ 60 crore BDT** | | | |


Project Budget Managemnet
Total: 60 crore BDT (approximately)
Maintenance (Post Project) 4%
Design and Planning 3%
Risk Management and Utility Conflict Resolution 17%
Surveying and Assesment 2%
Supervision and Monitoring Coordination 5%
Material Procurement 35%
Construction and Labour 34%
Surveying and Assesment
Design and Planning
Material Procurement
Construction and Labour
Supervision and Monitoring Coordination
Risk Management and Utility Conflict Resolution
Maintenance (Post Project)

## Project Timeline:

The estimated time for the project is around ***10 months*** provided that there would be no delay or unprecedented situation. After the project is completed, there would a timeline for ***2 years*** for post project management.

| Task | Timeline | Duration |
|---|---|---|
| Surveying and Assessment | January 1, 2026 – March 1, 2026 | 2 months |
| Design and Planning | March 1, 2026 – May 1, 2026 | 2 months |
| Material Procurement | May 1, 2026 – November 1, 2026 | 6 months |
| Construction and Labour | May 1, 2026 – November 1, 2026 | 6 months |
| Risk Management (Only Utility Conflict Resolution such as shifting gas pipes and electric lines) | May 1, 2026 – June 1, 2026 | 1 month |
| Supervision and Monitoring Coordination | January 1, 2026 – November 1, 2026 | 10 months |
| ***Total: 10 months (January 1, 2026 – November 1, 2026)*** | | |
| Maintenance (Post Project) | November 1, 2026 – November 1, 2028 | 2 years |

## Project Management Team Overview

The project will be led by a Project Manager, who serves as the primary decision-maker and liaison between BUET authorities, contractors, and stakeholders. Supporting the Project Manager are five core teams:

- **Consultants** – Provide expert advice on drainage, hydrology, and environmental compliance.
- **Civil Engineers** – Handle design, modeling, and technical supervision.
- **Construction Crew** – Execute on-site work, including excavation, pipe laying, and road resurfacing.
- **Technicians & Supervisors** – Oversee equipment, inspections, and daily progress.
- **Finance & Admin Team** – Manage budgets, procurement, and reporting.

  Each team has clearly defined tasks and accountability to ensure smooth project progression.

## Task Matrix

For the BUET Drainage System Improvement Plan, we've created a clear roadmap showing exactly which team handles which part of the project.

| Function/Role | Main Tasks | Key Responsibilities |
|---|---|---|
| Project Manager | Overall planning, scheduling & resources allocation | Lead teams, manage risks, stakeholder coordination |
| Consultants | Feasibility studies, design validation | Provide expert advice, ensure technical soundness |
| Civil Engineers | Survey, design, modeling (GIS), planning | Prepare detailed layouts, supervise execution |
| Construction Crew | Excavation, pipe laying, manhole installation, road resurfacing | Implement physical works safely & efficiently |
| Technicians & Supervisors | Equipment operation, routine inspections | Ensure machinery uptime, daily progress checks |
| Admin & Finance | Budget tracking, reporting | Control spending, process payments |

## Roles and Responsibilities by Phase

A well-structured chart of roles and responsibilities is fundamental to the successful implementation of the BUET Drainage System Improvement Plan.

| Role/Team | Survey & Assessment | Design & Planning | Procurement | Construction | Supervision & Monitoring | Cost Control & Reporting | Post-Project Maintenance |
|---|---|---|---|---|---|---|---|
| Project Manager | A | A | A | A | A | A | I |
| Consultants | R | R | C | I | C | I | I |
| Civil & Hydraulic Engineers | C | R | C | C | C | I | I |
| Construction Crew | I | I | I | R | C | I | I |
| Technicians & Supervisors | C | I | I | C | R | I | I |
| Financial Advisors | I | C | R | I | I | R | I |
| Admin & Coordination Team | I | I | C | I | C | C | I |
| BUET Authority & Sponsors | I | I | I | I | I | I | I |

## Project Quality Management

Project Quality Management ensures that all project deliverables meet defined standards and stakeholder expectations. It's a continuous process of planning, assuring, and controlling quality throughout the project lifecycle to prevent defects and ensure the outcome is fit for purpose.

The following table details our specific quality management framework, broken down by project phase and key quality areas:

- **Project Phase / Quality Area:** Defines the specific stage or aspect of the project being addressed.
- **Key Quality Objectives:** States the primary quality goals for that particular phase or area.
- **Metrics & Standards:** Specifies the measurable criteria or benchmarks used to assess quality.
- **Tools & Methods:** Lists the techniques, instruments, or processes employed to achieve and monitor quality.
- **Review/Audit Frequency:** Indicates how often formal quality checks are conducted.
- **Responsible Parties:** Identifies the individuals or teams accountable for quality in that area.

**Corrective Action Protocol:** Outlines the steps taken when quality issues are identified.

| Project Phase / Quality Area | Key Quality Objectives | Metrics & Standards | Tools & Methods | Review/Audit Frequency | Responsible Parties | Corrective Action Protocol |
|---|---|---|---|---|---|---|
| I. Surveying & Assessment | Data Accuracy & Completeness | Topographic survey deviation lepm5 cm Hydrologic data integrity 100% | GPS/Total Station Calibration GIS Data Validation Cross referencing with historical records | Post survey completion, Predesign approval | Civil Engineers Consultants | Re survey critical areas Data re validation & correction |
| II. Design & Planning | Design Efficiency & Compliance | Hydraulic capacity ge 150% of peak flow Adherence to national building codes Material specification accuracy 100% | HEC RAS Simulation Models Expert Peer Review Design Checklists | Design review meetings (monthly) Preconstruction audit | Consultants Civil Engineers | Design revisions & re simulations Additional expert consultation |
| III. Material Procurement | Material Quality & Specification | Concrete pipe strength (MPa) per IS/ASTM Cement grade verification (e.g., CEM I 42.5N) Supplier compliance rate > 95% | Independent Lab Testing Supplier Audits Delivery Inspection Checklists | Upon material delivery Random batch testing (weekly) | Procurement Team Financial Advisors | Reject non compliant materials Source alternative suppliers |

| | | | | | | |
|---|---|---|---|---|---|---|
| IV. Construction & Installation | Workmanship & Structural Integrity | Pipe alignment deviation lepm1 cm/m Compaction density ge 95% Proctor Road resurfacing smoothness (IRI le2.0 m/km) | On site Quality Checklists Daily Progress Reports Photographic Documentation Laser Alignment Tools | Daily site inspections Weekly progress reviews<2 br>Milestone completion audits | Technicians & Supervisors Construction Crew Civil Engineers | Rework non compliant sections Re installation of misaligned pipes Immediate repair of defects |
| V. Post Project Maintenance | System Functionality & Longevity | Drainage flow rate maintained at 90% capacity Blockage incidence reduction > 80% Structural integrity maintained (no cracks/subsidence) | Scheduled Cleaning Logs Performance Monitoring Sensors Visual Inspections & Surveys | Bi annually (first 2 years) Post heavy rainfall checks | Post Project Team | Immediate repair of damages Adjust maintenance frequency/methods Investigate recurring issues |

## Risk Identification & Management

The Risk Identification & Management section captures six major risk categories—Technical, Schedule, Resource, Operational, and Administrative—and assigns a severity score (Expectation × Impact) to each.

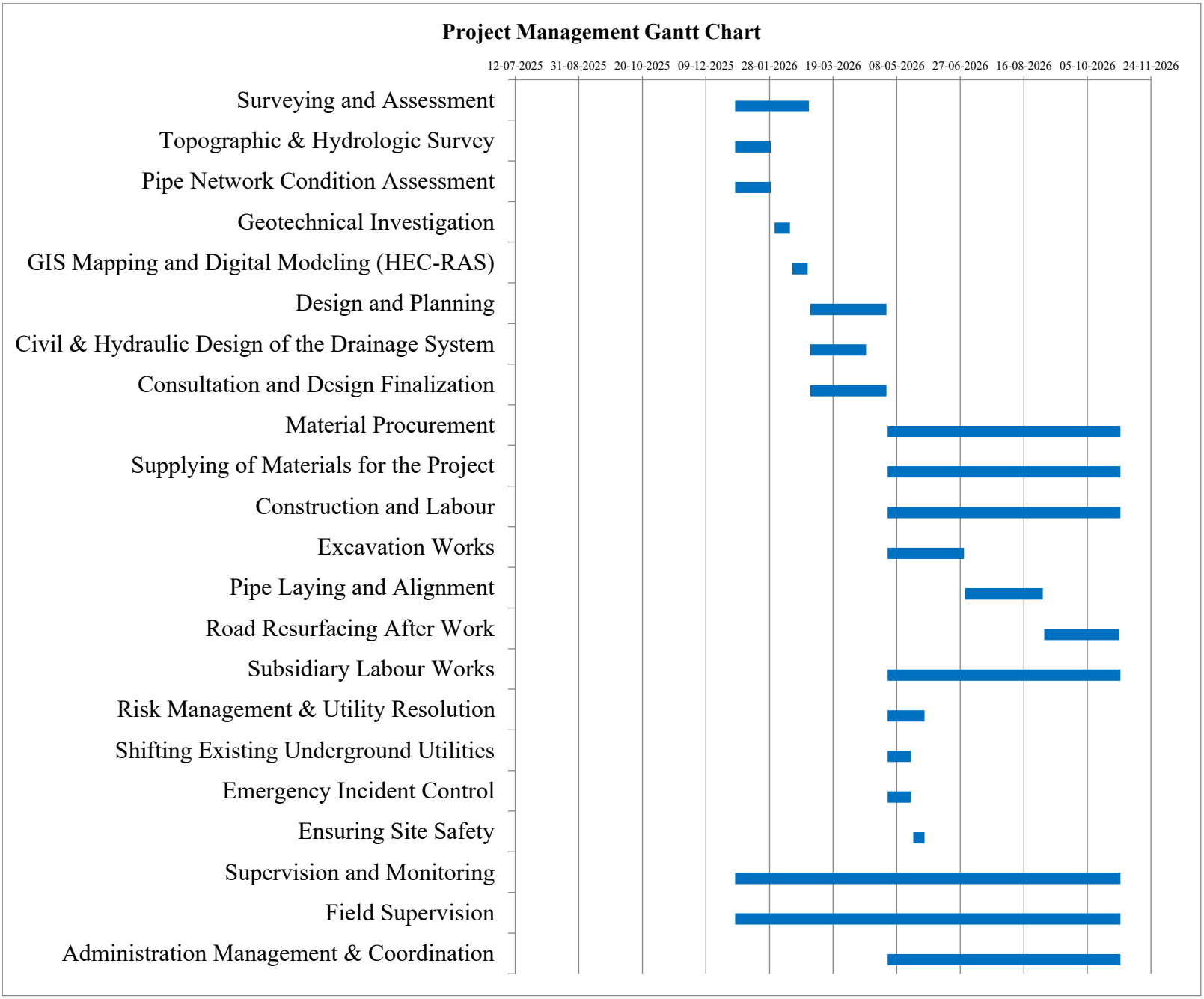
| Type of Risk | Jeopardy | Description of the Risk | Expectation of the risk (1-10) | Impact (1–10) | Severity of the risk (E × I) | Plan of Action |
|---|---|---|---|---|---|---|
| Structural challenges from retrofitting around aged buildings | Technical | Weak foundations. | 6 | 8 | 48 | Early structural survey; buffer design. |
| No existing bypass drainage during construction | Schedule | Flooding delays | 7 | 9 | 63 | Install temporary pumps/lines. |
| Utility conflicts with buried gas, water, electrical lines | Technical | Uncharted services | 8 | 7 | 56 | GPR survey; trial pits; utility coord. |
| Skilled labor shortage in specialized pipeline installations | Resource | Few specialist crews | 5 | 6 | 30 | Partner training; subcontract expert. |
| Student movement disruption during academic sessions | Operational | Access & noise impacts | 6 | 5 | 30 | Night/break work; signage; notices. |
| Delays from coordination with multiple authorities | Administrative | Slow approvals | 7 | 6 | 42 | Liaison officer. |

**Key Insights:**

- **Highest Risk (63):** Flooding delays due to lack of bypass drainage warrant immediate installation of temporary systems before peak rains.
- **Technical Risks:** Two technical items (structural retrofits and utility conflicts) score ≥48; early surveys and coordination are essential.
- **Operational & Resource Risks:** Although medium severity (30), these can disrupt progress if unaddressed. Training programs and optimized schedules will keep the campus functioning smoothly.
- **Administrative Coordination:** A liaison role is critical to reduce bureaucratic lag and keep milestones on track.

## Project Management Gantt Chart

The following Gantt chart summarizes the six major work streams and their constituent tasks. All dates refer to calendar year 2026.



| Work Stream & Sub-Tasks | Start | End | Duration |
|---|---|---|---|
| **A. Surveying & Assessment** | **Jan 1** | **Mar 1** | 2 months |
| • Topographic & hydrologic survey | Jan 1 | Jan 31 | 1 month |
| • Pipe-network condition assessment | Jan 1 | Jan 31 | 1 month |
| • Geotechnical investigation | Feb 1 | Feb 14 | 2 weeks |
| • GIS mapping & digital modeling (HEC-RAS) | Feb 15 | Feb 28 | 2 weeks |
| **B. Design & Planning** | **Mar 1** | **May 1** | 2 months |
| • Civil & hydraulic design of drainage system | Mar 1 | Apr 15 | 1½ months |
| • Consultation & design finalization | Mar 1 | May 1 | 2 months |
| **C. Material Procurement** | **May 1** | **Nov 1** | 6 months |
| • Ordering & supplying pipes, manholes, fittings & accessories | May 1 | Nov 1 | 6 months |
| **D. Construction & Labour** | **May 1** | **Nov 1** | 6 months |
| • Excavation works | May 1 | Jun 30 | 2 months |
| • Pipe laying & alignment | Jul 1 | Aug 31 | 2 months |
| • Road resurfacing after pipeline works | Sep 1 | Nov 1 | 2 months |
| • Subsidiary labor works (shoring, backfill, reinstatement) | May 1 | Nov 1 | 6 months |
| **E. Risk Managements & Utility Conflict Resolution** | **May 1** | **Jun 1** | 1 month |
| • Shifting existing underground utilities & emergency incident control | May 1 | May 20 | 20 days |
| • Ensuring site safety | May 21 | Jun 1 | 10 days |
| **F. Supervision & Monitoring** | **Jan 1** | **Nov 1** | 10 months |
| • Field supervision | Jan 1 | Nov 1 | 10 months |
| • Administration management & coordination | May 1 | Nov 1 | 6 months |

- **Surveying & Assessment** wraps up by March 1, giving the design team a full week to review survey deliverables before kickoff.

- **Design & Planning** runs from March 1–May 1, with 1½ months for engineering design and a 2-month concurrent window for stakeholder consultations.
- **Procurement** and **Construction & Labour** both span May 1–Nov 1; excavation (May–Jun), pipe works (Jul–Aug), and resurfacing (Sep–Nov) are sequenced to maintain a continuous workflow.
- **Risk Management & Utility Conflict Resolution** is fully contained within May 1–Jun 1, ensuring all safety and utility-relocation tasks are closed out before intensive pipe-laying begins.
- **Supervision & Monitoring** provides oversight from project start to finish, with administrative coordination ramping up at the same time as procurement and construction.

## Conclusion:

This project was undertaken with the aim of improving the drainage system of the BUET campus in a sustainable and practical way. It begins by addressing the core problem — how outdated, poorly maintained drainage is negatively affecting campus life, especially during the monsoon season. Daily routines are disrupted, and in some cases, health and safety are put at risk.

It is about finding a simple but effective way to improve drainage inside the BUET campus. Instead of just fixing broken pipes or clearing clogged drains temporarily, we're aiming to create a system that works better in the long run — especially during heavy rain and the monsoon season. The expected outcome is a cleaner, safer, and more functional campus environment where students and staff can move freely and work without interruption, even during heavy rain.

This is a non-commercial, student-led project developed solely for the benefit of the BUET community. Funding is expected from UGC and government sources, ensuring that the project remains both accessible and focused on public good.

This project represents a meaningful step toward creating a more resilient and sustainable campus environment. Together, we can build a campus that supports learning, health, and wellbeing for generations to come.